NATHAN RAWIRI

MECHANICAL & AEROSPACE ENGINEER

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Portfolio: rawnatural.github.io

PROFESSIONAL SUMMARY

Mechanical and aerospace engineer with 4+ years of hands-on design and manufacturing experience, proven project management capability, and strong CAD/FEA skills. Background spans precision machining and composite fabrication (aerospace rocketry), $50M railway infrastructure project management (Siemens Mobility), and supervision of advanced manufacturing facilities. Proficient in Autodesk Inventor and Siemens NX with practical FEA experience. Comfortable working independently, building client relationships, and adapting quickly to new project environments. Experienced operating lathes, mills, press brakes, water jets, and laser cutters. NZ citizen, available immediately.

EDUCATION

Bachelor of Engineering (Honours) - Mechanical and Aerospace Engineering

University of Queensland, Brisbane | 2021-2025 | GPA: 5.68/7.0

Aerospace Engineering Exchange Semester

TU Delft, Netherlands | 2024 | Composite structures, vehicle dynamics, autonomous systems

PROFESSIONAL EXPERIENCE

Project Engineer | **Siemens Mobility** | Wellington, New Zealand | 2023 (9 months)

- Managed $50M railway signalling modernisation project across Wellington rail network, coordinating 10-member multi-disciplinary engineering team (electrical, mechanical, civil, systems)
- Replaced legacy mechanical interlocking system with Westrace MkII and ETCS (European Train Control System), improving railway safety, reliability, and operational efficiency
- Generated technical installation requirements for 168 axle counters with precise positioning, mounting specifications, and site-specific hazard assessments
- Managed bill of materials aligned with engineering designs, procurement schedules, and installation sequences; ran weekly procurement meetings
- Procured $2M+ of specialised signalling equipment, managing vendor relationships and delivery schedules against project deadlines
- Chaired project meetings with internal teams, contractors, and client (KiwiRail); wrote and sent RFIs directly to client
- Led close-out of stalled post-commissioned project that had defeated multiple engineers over 4 years - reconstructed project history from scratch, coordinated all parties, and resolved every outstanding item before end of placement

Aerostructures Engineer & Deputy Project Lead | **UQ Space** | Brisbane, Australia | 2021-2025

- De facto aerostructures team lead for 6-member team within 40+ person engineering organisation, specialising in aluminium and composite fabrication for competitive rocket and rover programmes
- Designed and manufactured 3 complete rocket airframes using fiberglass/epoxy composite with vacuum bagging on precision mandrels; developed novel two-part nosecone mould process
- Manufactured aluminium rover chassis using water jet cutter, press brake, and lathe - still operational 5+ years later
- Conducted FEA analysis in Siemens NX on 30,000ft rocket structural components, validating wind shear loading and optimising strength-to-weight ratio
- Managed design-to-manufacturing workflow for 3 concurrent rocket programmes (5,000ft, 10,000ft, 30,000ft altitude targets)

- Served as Deputy Project Lead for 2nd place finish at Australian Rover Challenge 2025, managing cross-functional team of 40+ engineering students
- 1st place - AURC 2025, 5,000ft Rocket Category

Manufacturing Supervisor | **UQ Innovate Makerspace** | Brisbane, Australia | 2025

- Supervised up to 60 engineering students simultaneously in advanced manufacturing facility, ensuring safe operation of industrial machinery
- Operated and maintained 7+ types of manufacturing equipment: manual lathe, manual mill, press brake, laser cutter, water jet cutter, horizontal/vertical band saws
- Provided hands-on technical support for aluminium and steel fabrication, precision machining, welding, and assembly operations
- Troubleshot complex mechanical assemblies and provided design-for-manufacturing guidance to improve part quality

Mechanical Systems Design Tutor | **University of Queensland** | Brisbane, Australia | 2025

- Tutored 250 mechanical engineering students in MECH3100, the flagship mechanical design and manufacturing course at UQ; one of only four tutors selected for hands-on manufacturing expertise
- Worked intensive 64-hour schedule over 5 weeks providing hands-on manufacturing support in makerspace facility
- Guided students through complete vehicle design and fabrication including transmission systems, drive mechanisms, and traction subsystems capable of towing 1kN loads
- Designed and built personal gearbox tractor project within $400 budget - designed entirely solo in Autodesk Inventor, manufactured all components

TECHNICAL SKILLS

CAD & Design	Autodesk Inventor (advanced), Siemens NX, assembly design, technical drawings, GD&T
Analysis	FEA (Siemens NX, ANSYS), StarCCM+ CFD, MATLAB, Simulink, structural & thermal analysis, ANSYS Maxwell
Manufacturing	Precision machining, sheet metal fabrication, composite layup (fiberglass/epoxy, vacuum bagging), welding, workshop safety management
Machine Tools	Manual lathe, manual mill, press brake, laser cutter, water jet cutter, band saws, drill press, grinder
Materials	Aluminium alloys, mild & stainless steel, composites (fiberglass/epoxy/carbon fibre), material selection
Project Mgmt	Procurement, BOMs, stakeholder coordination, schedule management, technical documentation, risk assessment
Software	Python, C++, MATLAB, JavaScript, PHP, iOS Swift (see GitHub portfolio)

KEY ACHIEVEMENTS

- 1st Place - AURC 2025, 5,000ft Rocket Category
- 2nd Place - Australian Rover Challenge 2025 (Deputy Project Lead, 40+ person team)
- Successfully managed $50M railway signalling infrastructure project at Siemens Mobility
- Led procurement of $2M+ specialised equipment with zero cost overruns
- GPA: 5.68/7.0 - Strong academic performance in Mechanical & Aerospace Engineering
- MENSA Member
- Half-Ironman Finisher (2023)

REFERENCES

Mark Kilby - Head of Project, Systems & Specialty Engineering, Siemens Mobility
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James Orman - Managing Director, UQ Space
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Oliver Pettingill - Former Aerostructures Lead, UQ Space (now Powertrain Engineer, Mercedes AMG F1)
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